

# Acceptance Testing

Spring, 2026

# Contents

- What is Acceptance Testing
- Contents of Acceptance Testing
- Alpha Testing & Beta Testing
- Brief comparison of four phases of software testing

# 1. What is Acceptance Testing

- Acceptance testing is the **final level of software testing** where the system is tested for **compliance to its business requirements**.
- It is performed by the client or the end users with the intent to see if the product is fit for delivery.
- It is carried out after system testing and before the final delivery to the client.
- It works under the **Black Box Testing Method**.

## 2. Contents of Acceptance Testing

Two parts: Software Configuration Review and Software Validity Testing.

### 1. Software configuration review

Common software configuration items include :

- (1) Main **software program** configuration, generally including source program, executable program, software installation and configuration script, key test script or test program
- (2) Main **technical documents**
- (3) Main **development management documents**

# Contents of Acceptance Testing

➤ Complete the following checks for the software configuration :

(1) **Source code check**

- Normative check
- Data type check
- Check external interfaces

(2) **Software consistency check**

- Compile check
- Install and uninstall tests
- Run the module consistency check

➤ Check the following items for documents :

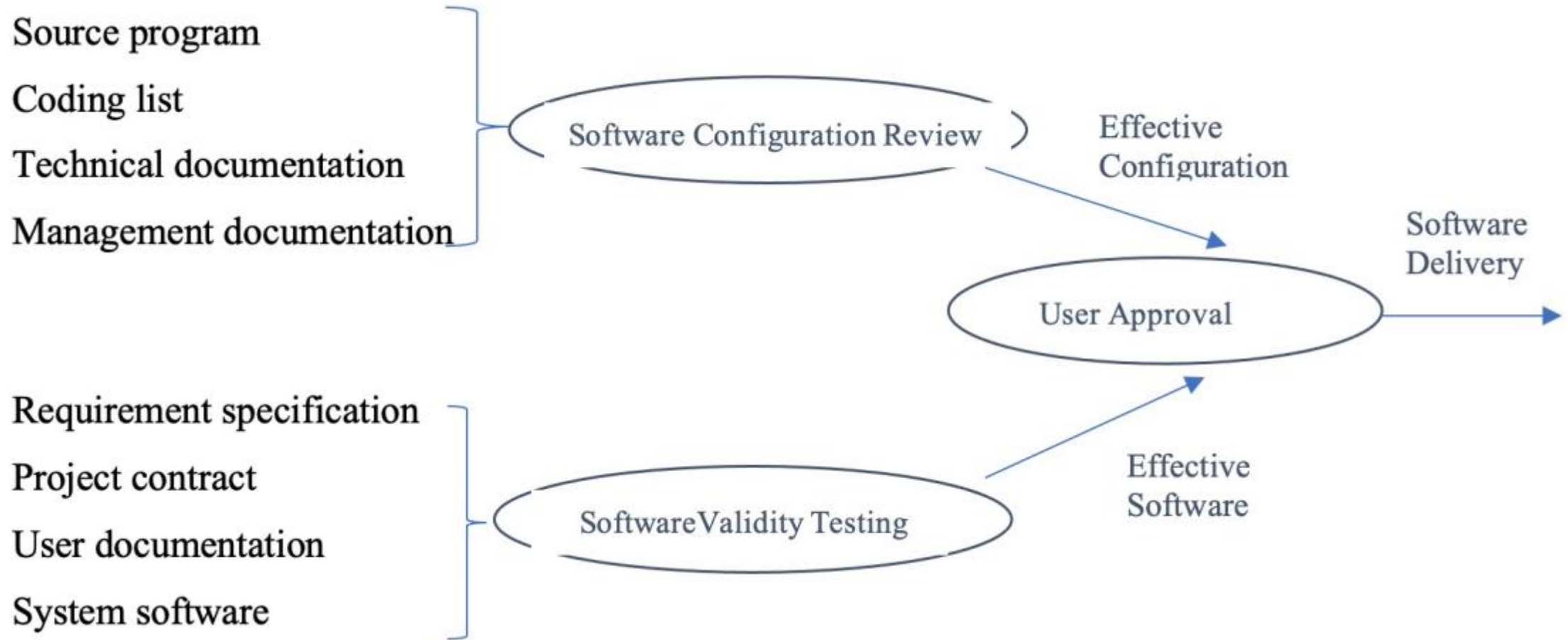
- Completeness
- Normative
- Targeted
- Complete independence
- Flexibility
- Traceability

# Contents of Acceptance Testing

## 2. Software validity test

- Software interface testing
- Usability testing
- Functional testing (including normal business process testing and error handling capability testing)
- Performance test (including load, volume and stress testing)
- Software executing environment and system platform configuration testing
- Robustness testing( including recovery tests under various hardware and software failures)
- Reliability testing
- Compatibility testing
- Data backup testing
- Security testing

# Contents of Acceptance Testing



## Acceptance Testing Consideration

- (1) Before the acceptance test, a **formal acceptance test plan** shall be prepared to clarify the **criteria** for passing the acceptance test, which shall be confirmed by users.
- (2) The acceptance test must be carried out in the **actual use environment** of end users, or **simulate the actual operation environment** of users, so as to avoid some potential problems of software that cannot be found due to environmental differences.
- (3) Acceptance tests should cover **coarse-grained, business-level functions** of the software. There is **traceability** between acceptance test cases and software project contracts and software requirements specifications.

It is not possible or necessary for acceptance test cases to re-run all test cases conducted during the development phase.



## Acceptance Testing Consideration

(4) The acceptance test must be **user-oriented** and conducted in a way that users can **intuitively** perceive from the perspective of the actual business scenarios used by end users.

(5) The design of acceptance test cases must fully consider the **user's way of thinking**, usage habits, business language, etc., and organize test cases and test procedures according to the **main business scenarios**. It focuses on the function points and performance points that customers care most about.

### 3. Alpha Testing & Beta Testing

- **Alpha and Beta testing** are Acceptance Testing types (Customer Validation methodologies ) that help in building confidence to launch the product, and thereby result in the success of the product in the market.
- Alpha and Beta Testing phases mainly focus on discovering the bugs from an already tested product and they give a clear picture of how the product is used by the real-time users.

## Alpha Testing

- Alpha testing is the form of acceptance testing that takes place **at the developer's site**.
- It can be carried out by both in-house developers and QAs as well as potential end-users as well.
- Alpha testing is **not open** to the world.

## Beta Testing

- Beta Testing (Field Testing) is the form of acceptance testing that takes place **at the customer's or the end user's site.**
- It is performed after alpha testing and **in the real-world environment** without the presence or control of developers.
- Beta tests or the **beta version** of the application are normally open to the whole world (or client).

**For example,** recently Microsoft corporation released Windows 10 beta and based on the feedback from thousands of users they managed to release a stable OS version. In the past, Apple also released OS X beta in public and fixed many minor issues and improved the OS based on user feedback.

# Alpha Testing & Beta Testing

## Alpha Testing

## Beta Testing

### Basic Understanding

First phase of testing in Customer Validation

Second phase of testing in Customer Validation

Performed at developer's site - testing environment. Hence, the activities can be controlled

Performed in real environment, and hence activities cannot be controlled

Build released for Alpha Testing is called Alpha Release

Build released for Beta Testing is called Beta Release

Issues / Bugs are logged into the identified tool directly and are fixed by developer at high priority

Issues / Bugs are collected from real users in the form of suggestions / feedbacks and are considered as improvements for future releases.

# Alpha Testing & Beta Testing

## Participants

Technical Experts, Specialized Testers with good domain knowledge (new or who were already part of System Testing phase), Subject Matter Expertise

End users to whom the product is designed

## Test Duration

Many test cycles conducted

Only 1 or 2 test cycles conducted

## Rewards

No specific rewards or prizes for participants

Participants are rewarded

## 4. Brief comparison of software testing phases

| Items        | Unit Testing                                                    | Integration Testing                                                       | System Testing                                       | Acceptance Testing                                        |
|--------------|-----------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| Test object  | Software units, such as functions, classes, components, modules | Interfaces between modules, such as parameter passing                     | The entire system, including hardware and software   | The entire system, including hardware and software        |
| Test basis   | Detailed software design                                        | Software architecture design                                              | Software requirements specification                  | Requirements specification, contract, acceptance criteria |
| Tester       | Developer or white box test engineer                            | Developers and testers work together                                      | primarily the responsibility of professional testers | User-dominant, developer and tester work together         |
| Test methods | White box test is mainly used                                   | Black box testing is mainly used while white box testing is the secondary | Fully black-box testing                              | Fully black-box testing                                   |
| Test data    | Real data is generally not used                                 | Real data is generally not used                                           | Use or simulate real business data whenever possible | Use or simulate real business data whenever possible      |